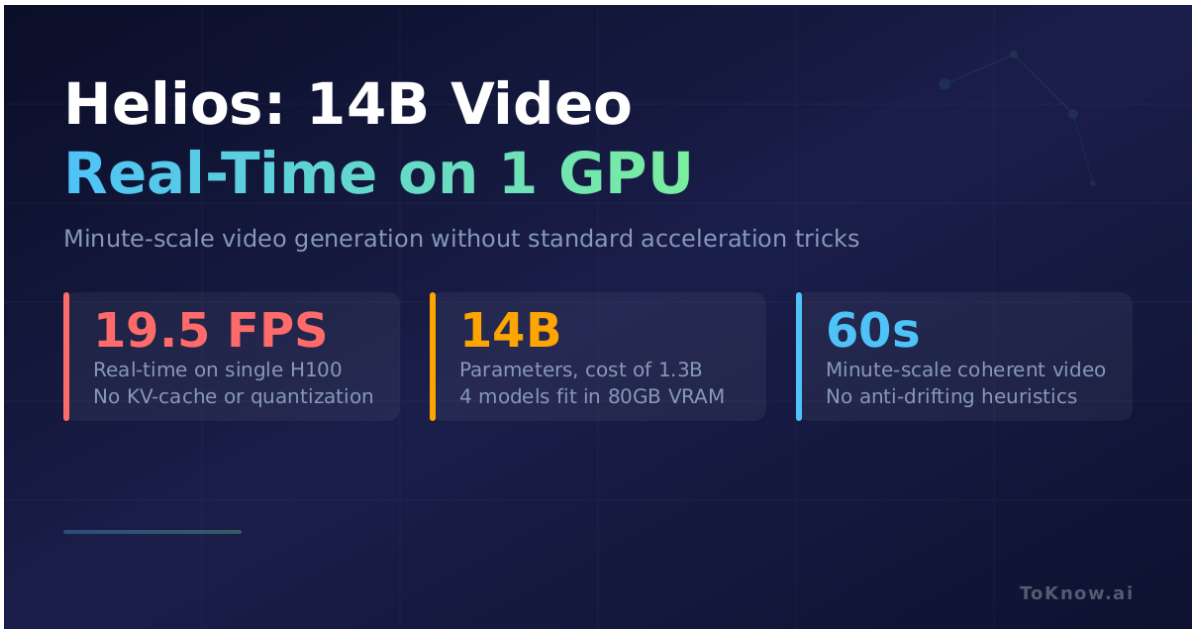
Helios: A 14B Video Model That Runs at 19.5 FPS on a Single GPU

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Researchers at ByteDance and Peking University released Helios, a 14-billion parameter autoregressive diffusion model that generates video at 19.5 frames per second on a single NVIDIA H100 GPU. The model produces minute-scale clips (up to 60 seconds at roughly 24 FPS output) and natively handles text-to-video, image-to-video, and video-to-video tasks through a unified input representation. What makes Helios unusual is what it does not use: there is no

KV-cache, no sparse or linear attention, no quantization, and no anti-drifting heuristics like self-forcing or keyframe sampling. Instead, it compresses historical and noisy context aggressively and reduces sampling from 50 steps to 3 via adversarial hierarchical distillation. The result is inference costs comparable to models one-tenth its size. Training is similarly lean: four 14B models fit within 80 GB of GPU memory without parallelism or sharding frameworks.

That speed changes what video generation can be used for. Previous models at this quality level took minutes to produce seconds of footage, limiting them to offline workflows. Helios at real-time speeds opens the door to interactive applications: live content creation tools, game engines fed by generative video, and on-the-fly storyboarding. The code, base model, and distilled model are all released under Apache 2.0 with day-one support from Diffusers, vLLM-Omni, and SGLang.

This is ByteDance's second major video generation release in recent weeks, following Seedance 2.0's unified multimodal approach. Where Seedance focused on multi-input control and audio-video fusion, Helios targets raw generation speed and long-form coherence, two problems the field has treated as fundamentally at odds with scale.

Sources:

- Helios: Real Real-Time Long Video Generation Model (arXiv)
- Helios GitHub Repository
- Helios on Hugging Face Papers
- Helios Project Page

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